

Transformation in network and edge platform

The fixed-function solutions that were replaced by the newest programmable platform for network and edge workloads are the infrastructure processor unit, or the IPU. It is helpful for people who are operating the world's largest cloud data centres and cloud service providers.

The CPUs are used for two different purposes. One, CPUs run the software infrastructure that keeps the data centre running, and two, a larger set of CPUs run the tenant's workloads. It is extremely important that the tenant workloads, over which the cloud providers have no control, don't disrupt either the infrastructure itself or the other tenants. Presently, it is hard to protect one from the other if they both run on the same CPU cores.

The job of the IPU is to run the infrastructure workloads on a separate, secure, and isolated set of CPU cores. Now, the IPU is versatile and programmable. It's a programmable platform of its own. Not only does it have an array of CPU cores, but it also has a packet processing pipeline that is programmable like Tofino using the p4 language as well as a variety of accelerators, such as encryption and compression, which are also available for the developer to program for themselves.

It is important for this new class of computing devices to be open and programmable as it helps com-

Newest Programmable Platform – the IPU

New Intelligent Data Center Architecture: Infrastructure Processing Unit (IPU)

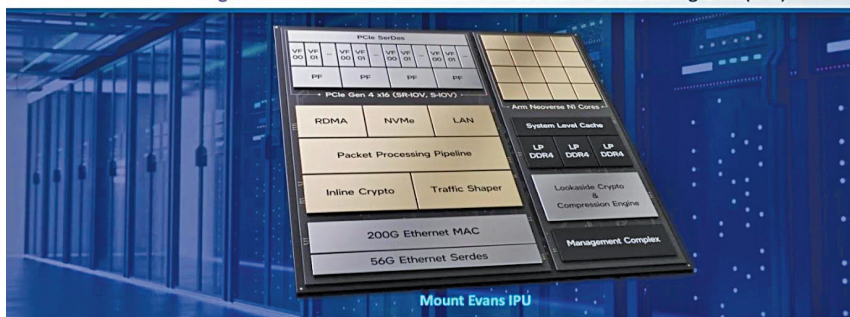


Fig. 2: IPU, the new data centre architecture

panies keep improving and involving their data centres without being locked into a single solution. In fact, by making the system open, the IPDK open development environment helps make the entire ecosystem bigger for the entire industry as a whole. The power of this programming framework will help you to develop a program with a description of the functionality and characteristics of a pipeline and compile it to whatever target is appropriate for the application.

For example, if you're stringing together a number of workloads, you can target a host CPU via DPDK. This flexibility and uniformity are made possible through compiler-friendly programmable silicon designing chips, with a program in mind, and allowing them to write their own p4 programs to determine the forwarding behaviour that they

need. This comes together under a common and open software framework that exposes open interfaces and attracts developers to invest in delivering platform solutions for a variety of verticals.

This can be deployed on whichever silicon platform is best suited for the particular vertical and use cases, whether it's in the data centre, the network, or the edge. Across vertical markets, customers are demanding the platforms, the frameworks, and the developer tools that combine compute, storage, and networking to empower them to consolidate workloads, bring data together with applications, and perform real-time analytics.

The interesting opportunity is to deliver programmable networks that span the cloud out to the edge, where the programmable elements work together to improve the network performance, latency, and reliability from end to end. This requires a deep understanding of the workloads that our customers are seeking to deploy and optimise, and an inability to distill that implementation down to its essence, so it can be used for a variety of different workloads.

The more the silicon designers understand the workloads and the use cases, the more they implement the design with the software developer in mind. The more they develop as a compiler target, the more likely

The four superpowers that are driving unprecedented semiconductor demand across the industry

- 1. Ubiquitous computing.** Everything we interact with involves computer technology.
- 2. Cloud-to-edge infrastructure.** There is an immense amount of computer communication between servers in the cloud and data centres, and they demand the highest performance communication speeds in the world. The network is moving to software to allow new services to be introduced in software, resulting in an increasingly intelligent edge.
- 3. Pervasive connectivity.** Everyone and everything is linked.
- 4. Artificial intelligence.** Attempting to make sense of the importance of an infinite amount of data.

Each superpower reinforces and amplifies the others to help customers capitalise on this fastest-growing opportunity.